



GIA - IAA

Introducció a

l'Aprenentatge Automàtic

1. Laboratory: Introduction (I)

Jordi Luque
jordi.luque.serrano@upc.edu

Evaluació

- ❖ Teoria
- ❖ Laboratori
- ❖ La assignatura consta d'un exàmen parcial (P) y un final (F).
- ❖ El laboratori serà evaluat de manera continua (LC) i amb una entrega (LF).
- ❖ $\text{Nota Final} = (0.2 * P) + (0.4 * F) + \mathbf{(0.1 * LC) + (0.3 * LF)}$
 - **Entrega Final (LF): La setmana del 11 de novembre** es farà el lliurament i explicació de l'enunciat a classe de laboratori. **Entrega abans del 23 de desembre.**

Temari

Primer bloc dedicat a l'anàlisi i tractament de les dades:

- ❖ Introducció a l'Aprenentatge Automàtic.
- ❖ Disseny d'experiments en aprenentatge automàtic.
- ❖ Preprocessament de dades
- ❖ Regressió aplicada
- ❖ Reducció de dimensionalitat

Segon bloc dedicat a algorismes de classificació supervisats:

- ❖ Classificació: Conceptes previs i revisió de mètodes bàsics
- ❖ Mètodes de classificació basats en altres criteris..
- ❖ Multiclassificació

Tercer bloc dedicat a algorismes de classificació no supervisats i altres com ètica,

- ❖ Explicabilitat
- ❖ Clustering
- ❖ Algorismes Grafs

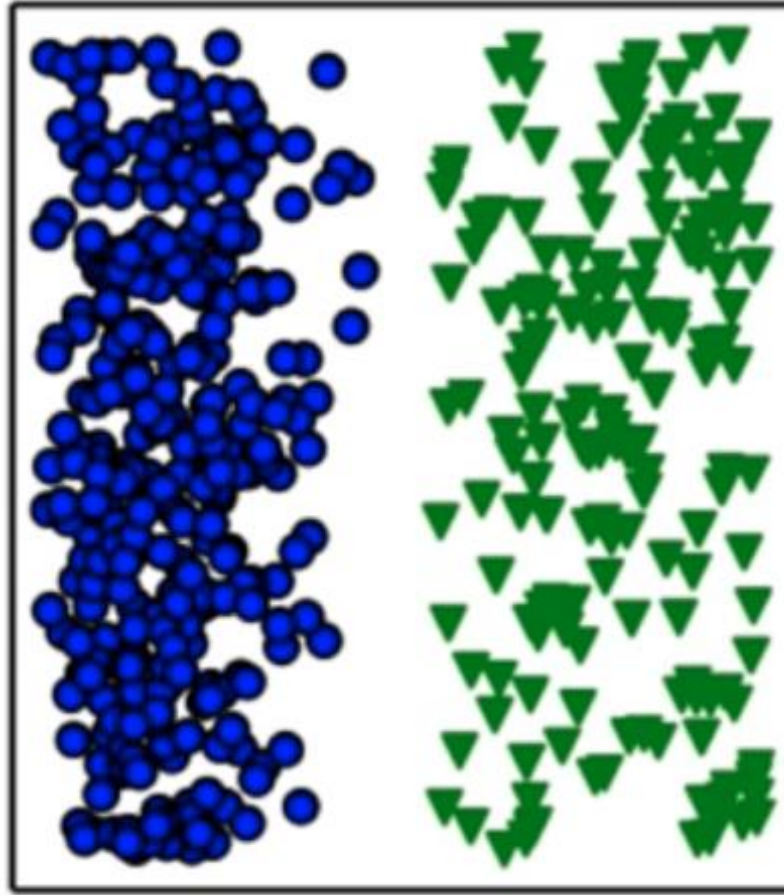
Bibliography

- ❖ *Bishop, Christopher M. Pattern recognition and machine learning. New York: Springer, cop. 2006. ISBN 0387310738*

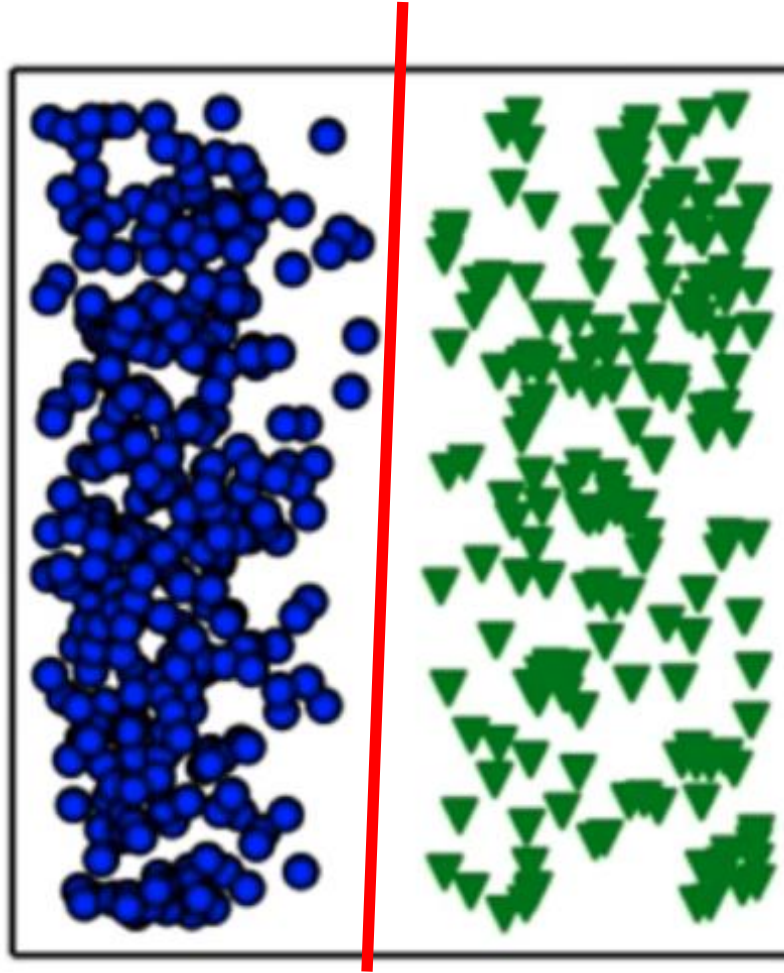
Other nice references:

- ❖ *Mark Coeckelbergh, Ética de la inteligencia artificial. Editorial Cátedra. ISBN: 978-8437642123*
- ❖ *Bradley Efron and Trevor Hastie, Computer Age Statistical Inference: Algorithms, Evidence and Data Science, Cambridge University Press, Student Edition 2021, ISBN: 978-1108823418*
- ❖ *Sebastian Raschka and Vahid Mirjalili, Python Machine Learning - Third Edition: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2, Packt Publishing 2020, ISBN: 978-1789955750*
- ❖ *Trevor Hastie and Robert Tibshirani and Jerome Friedman. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Second Edition (Springer Series in Statistics) 2nd Edition, 2016, ISBN: 978-0387848570*

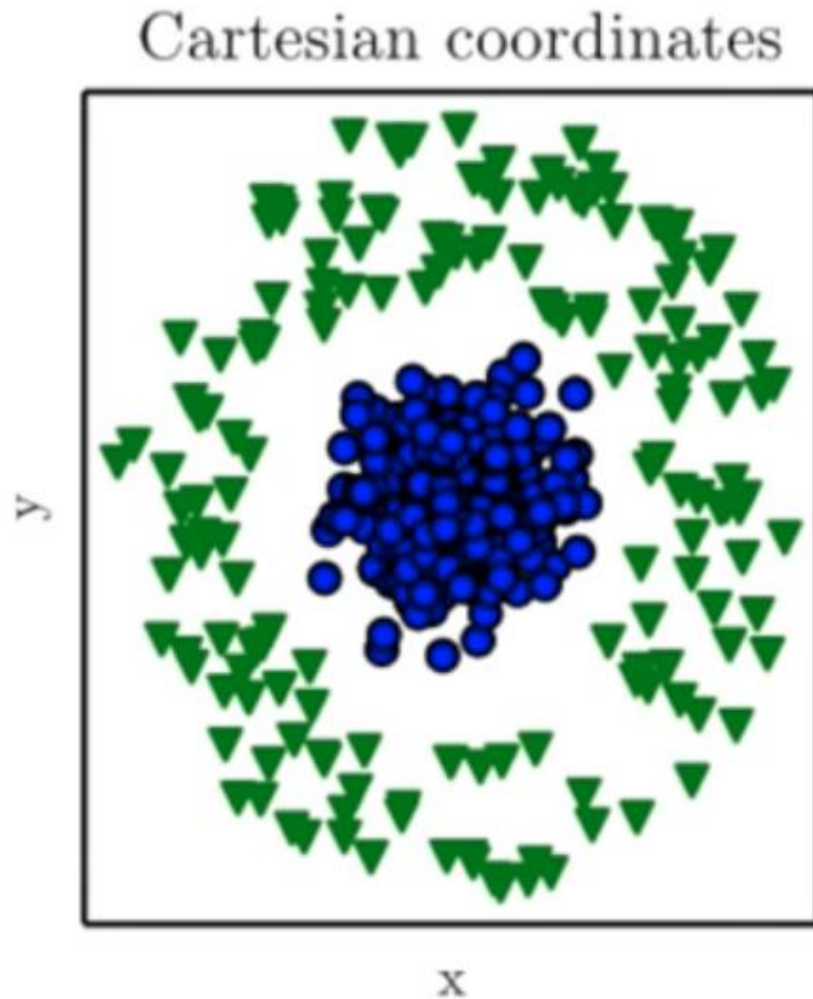
Veure les dades



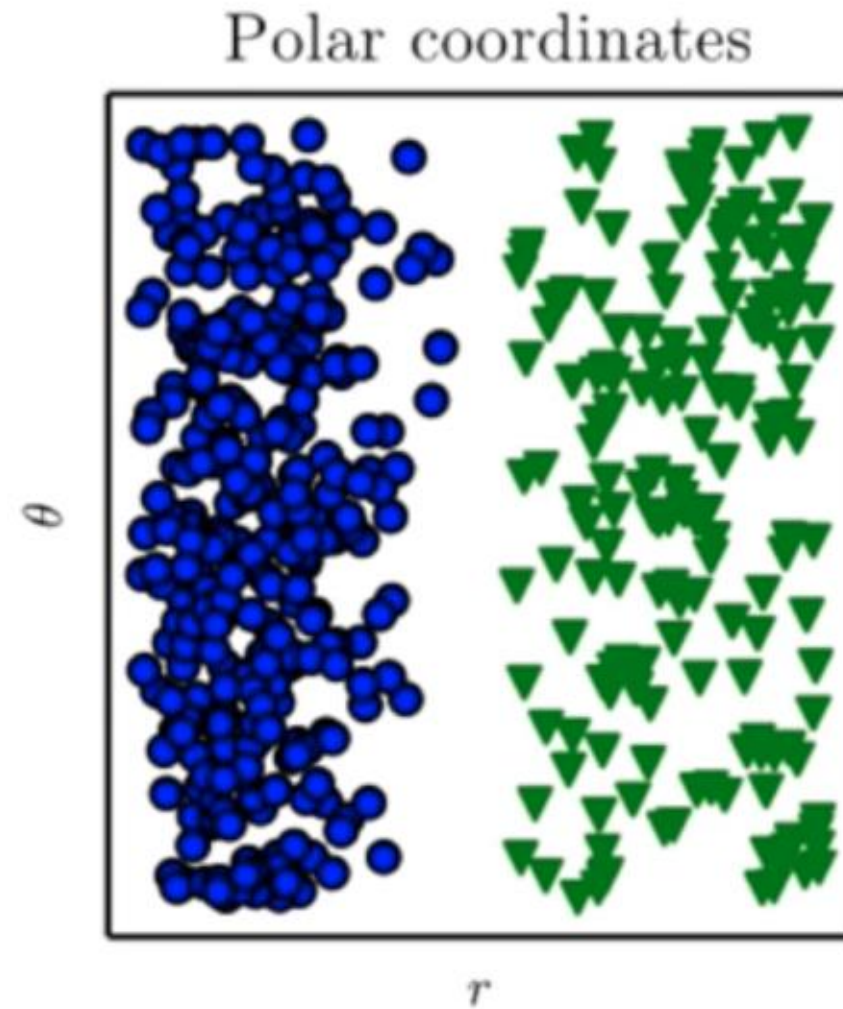
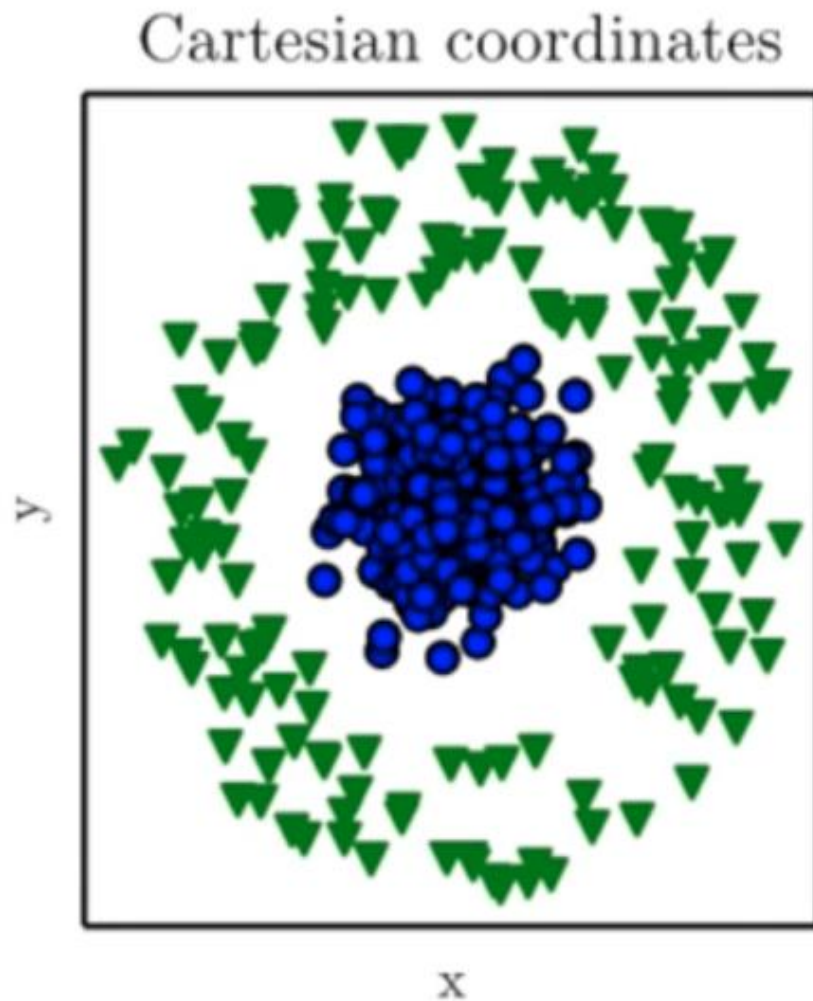
Separació de les classes



Representació de les dades és important



La representació de les dades és important



Python

- ❖ Editor: Visual Studio Code <https://visualstudio.microsoft.com/downloads/>

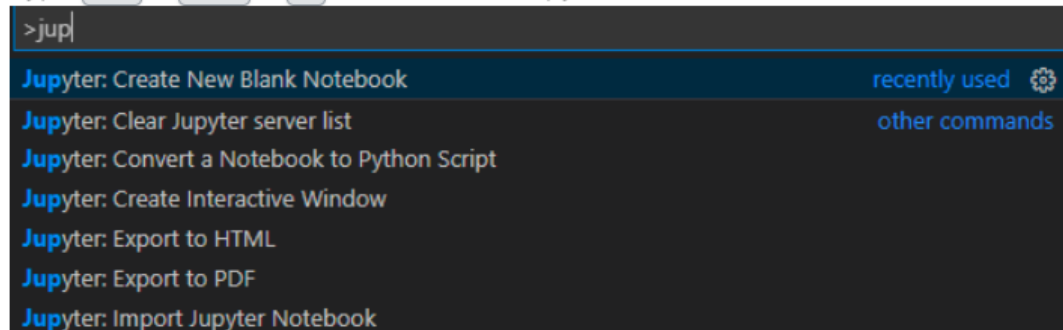


- ❖ Llenguatge més usats python i R
- ❖ Gestió d'entorns de python (opcional)
 - pyenv, <https://github.com/pyenv/pyenv> (veure README)
 - Linux | Mac podeu fer servir la instal·lació automàtica: `curl https://pyenv.run | bash`

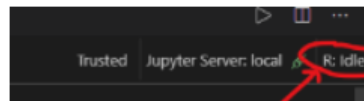
Basics: editor

- ❖ Visual Studio Code <https://visualstudio.microsoft.com/downloads/>

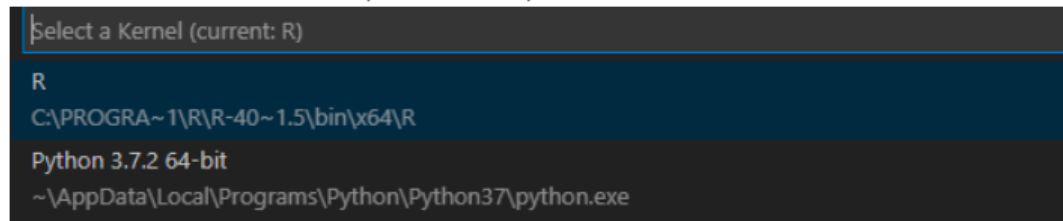
1. Reload Window `Ctrl + R`
2. Type `Ctrl + Shift + P` to search for "Jupyter: Create New Blank Notebook"



3. Click on the button right below ellipsis in upper right corner to choose kernel



4. Switch to the desired kernel, in this case, R's



Basics: Instal·lació de llibreries

```
1  %%matplotlib notebook
2  import numpy as np
3  import matplotlib.pyplot as plt
4  import seaborn as sns
5  import pandas as pd
6  from collections import Counter
7  from IPython.core.interactiveshell import InteractiveShell
8  import warnings
9
10
11  InteractiveShell.ast_node_interactivity = "all"
12  #pd.set_option('precision', 3)
13  display.precision = 3
14  sns.set()
15  warnings.filterwarnings('ignore')
```

[1] ✓ 9.4s Python

```
1  # extra imports
2  from pandas import read_csv
3  from sklearn.neighbors import KNeighborsClassifier
4  from sklearn.neighbors import LocalOutlierFactor
5  from sklearn import preprocessing
6  from statsmodels.genmod.generalized_linear_model import GLM
7  from pandas.plotting import scatter_matrix
8  from scipy.stats import boxcox
```

[2] ✓ 1.8s Python

Codi de laboratori

INSTRUCCIONS

No cal entregar res, cal assimilar

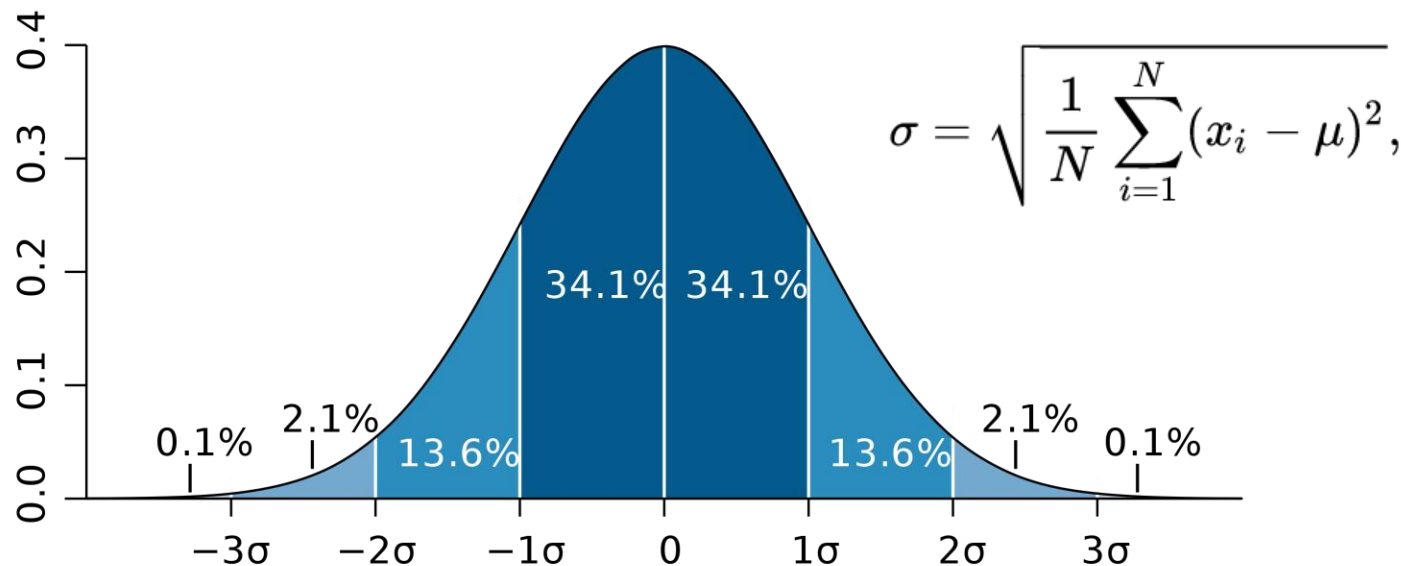
S'ha de fer amb calma, sense córrer, mirant d'entendre en tot moment què s'està fent **a nivell conceptual** i quin efecte té

Podeu deixar el codi python en si (crides, paràmetres, sintaxi) per una segona lectura o com a treball personal. El codi python és un mitjà, no un objectiu.

Algoritmes i inferencia

"Very broadly speaking, **algorithms** are what statisticians do while **inference** says why they do them. A particularly energetic brand of the statistical enterprise has flourished in the new century, data science, emphasizing **algorithmic thinking** rather than **its inferential justification**."

(Fonte: Efron, B., & Hastie, T. (2016). *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*. Cambridge: Cambridge University Press.)



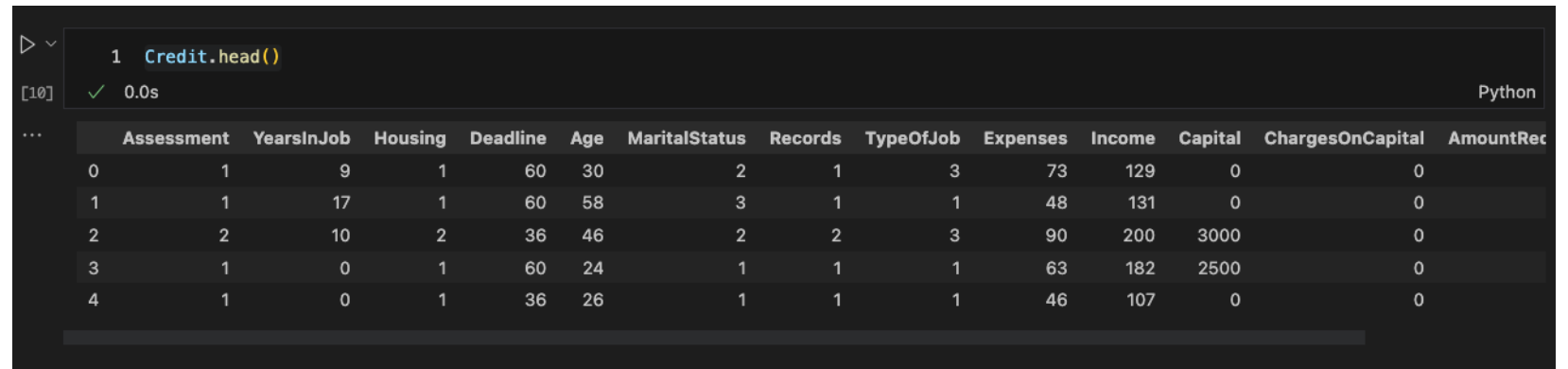
$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$$

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}.$$

$$SE \approx \hat{se} = \frac{s}{\sqrt{n}}$$

The data

- ❖ Three different datasets for the intro:
 - Dr. Myers' nephrology laboratory (kidney patients)
 - Credit scoring
 - Airline Passenger Satisfaction



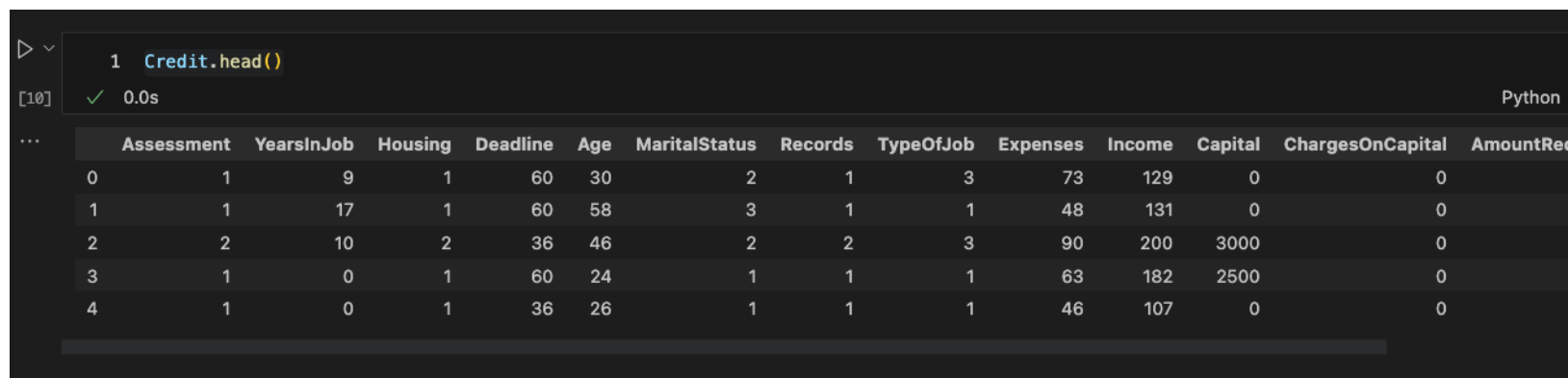
The screenshot shows a Jupyter Notebook interface. The code cell contains `1 Credit.head()`, which has been executed successfully, as indicated by a green checkmark and a runtime time of 0.0s. The output is a table with 14 columns and 5 rows of data.

	Assessment	YearsInJob	Housing	Deadline	Age	MaritalStatus	Records	TypeOfJob	Expenses	Income	Capital	ChargesOnCapital	AmountRec
0	1	9	1	60	30	2	1	3	73	129	0	0	
1	1	17	1	60	58	3	1	1	48	131	0	0	
2	2	10	2	36	46	2	2	3	90	200	3000	0	
3	1	0	1	60	24	1	1	1	63	182	2500	0	
4	1	0	1	36	26	1	1	1	46	107	0	0	

Reading a dataset

(function) `def read_csv( ... ) --> DataFrame`

- ❖ Reading properly a data set is non-trivial because you need to know
- ❖ Separator column separator, is there a header? how are strings quoted? how (if any) are missing values coded? should character vectors be converted to factors? Should white spaces be stripped?
- ❖ Read the file ``credsco.csv`` and inspect it. Add a cell code to display some information from the file using the `head()` function.



The screenshot shows a Jupyter Notebook interface. In the code cell, the command `1 Credit.head()` has been executed. Below the code, the output is displayed as a DataFrame with 14 columns and 5 rows of data. The columns are: Assessment, YearsInJob, Housing, Deadline, Age, MaritalStatus, Records, TypeOfJob, Expenses, Income, Capital, ChargesOnCapital, and AmountRec. The rows are indexed from 0 to 4.

	Assessment	YearsInJob	Housing	Deadline	Age	MaritalStatus	Records	TypeOfJob	Expenses	Income	Capital	ChargesOnCapital	AmountRec
0	1	9	1	60	30	2	1	3	73	129	0	0	
1	1	17	1	60	58	3	1	1	48	131	0	0	
2	2	10	2	36	46	2	2	3	90	200	3000	0	
3	1	0	1	60	24	1	1	1	63	182	2500	0	
4	1	0	1	36	26	1	1	1	46	107	0	0	

Pandas DataFrame

(function) `def read_csv( ... ) --> DataFrame`

- ❖ `Credit[:4]` indexes for 4 first samples (starting by 0 index), all columns
- ❖ `Credit.iloc[0,4:8]` index columns from 4th to 7th from the sample index 0
- ❖ `Credit.loc[0,'Age':'TypeOfJob']` index by column name
- ❖ Write a code for accessing last 10 samples and previous columns. Example of output:

PLAYTIME: Practice how to check other variables, and other samples (e.g., last 10?)

	Age	MaritalStatus	Records	TypeOfJob
4445	22	1	2	1
4446	50	2	2	3
4447	52	2	1	1
4448	49	2	1	1
4449	30	2	2	2
4450	39	2	1	1
4451	46	2	1	1
4452	37	2	1	2
4453	23	1	1	3
4454	32	2	1	3

Detecting outliers using IQR (Interquartile range)

- ❖ First quartile or Q1 is 25/75 data split
- ❖ Third quartile or Q3 75/25 data split
- ❖ One easy criteria to decide which values are outliers is to use the distance between quartiles. Specifically, you define the outliers as:
 - ❖ The values smaller than $Q1 - 1.5 \times IQR$
 - ❖ The values bigger than $Q3 + 1.5 \times IQR$

This is the criteria used in the boxplots to mark the outliers.

